

A Study on the System Strength Evaluation Index Considering the Interaction Between Renewable Energy

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Abstract—In order to reduce greenhouse gas emissions, the connection of renewable energy to power system is increasing. Renewable energy such as wind power and photovoltaic are inverter-based resources (IBRs) that provide limited short-circuit current unlike synchronous generators, and the limited short-circuit current adversely affects voltage after disturbance. Describes Short Circuit Ratio(SCR), a system strength evaluation index used in power systems when connecting inverter-based resources, and Weighted Short Circuit Ratio(WSCR) and Composite Short Circuit Ratio(CSCR), system strength evaluation indexes considering the interaction between inverter-based resources caused by Phased-Locked Loop(PLL) control characteristics is analyzed.

Keywords—System Strength, System Strength index, interaction between IBGs, Grid-following inverter, Grid-forming inverter

I. INTRODUCTION

Due to the international climate issues, interest in reducing greenhouse gases is increasing. In line with this trend, the power system is increasing the supply rate of renewable energy using wind power and photovoltaic to reduce greenhouse gas emissions. Renewable energy is a power generation source connected to the grid through an inverter, and the increase in grid connection of these inverter-based resources can adversely affect the stable operation of the grid. Existing synchronous generators provide short-circuit current of 7 p.u. of rated [1], whereas inverter-based resources provide a limited short-circuit current of 1.1 to 1.2 p.u. due to the mechanical characteristics of the inverter [2]. Due to the characteristics of these inverters, in areas where inverter-based resources are intensively connected, the strength of the grid is lowered, and the possibility of voltage instability after a major disturbance in the grid increases. Therefore, in the power system, system strength weak areas are determined through the system strength evaluation of the system connected to renewable resources, and countermeasures are devised. A typical system strength evaluation index is the SCR index. The SCR index is an indicator that determines strength by calculating the ratio of the installed capacity of the inverter-based resources to the SCC before connection in the bus connected to the inverter-based resource. If the SCR indicator is less than 3, it is evaluated as a very weak system [3-5].

Inverter-based resources can be divided into two types based on inverter facility control characteristics, which are divided into Grid-following(GFL) inverter and Grid-forming(GFM) inverter. Currently, the main inverter is the

GFL inverter, and the big difference from the GFM inverter is that the GFL inverter measures the voltage of the system through an Phase-locked Loop(PLL) and controls the phase angle. However, in areas where inverter-based resources are intensively connected, interactions between inverter-based resources are caused by the inner current loop of the GFL inverter, which adversely affects voltage wave control, resulting in adverse effects such as voltage oscillation. Therefore, by processing the SCR indicator, which is an indicator for evaluating the existing system strength, Electric Reliability Council of Texas(ERCOT) uses the Weighted Short Circuit Ratio(WSCR) index and General Electric(GE) uses the Composite Short Circuit Ratio(CSCR) index to reflect the interaction between inverter-based resources.

In this paper, the characteristic of GFL inverter is described and the system strength index considering the interaction between GFL inverter facilities is introduced.

II. GRID FOLLOWING INVERTER

A. Non-synchronous Machine Source

Non-synchronous machine source is largely classified into Grid-following and Grid-forming [6-7]. GFL is an adjustable current source and GFM is an adjustable voltage source in the power system. The largest difference between GFL and GFM is the presence or absence of PLL to maintain system synchronization. GFL can be subdivided into two types depending on whether it can support the bus voltage of the AC system. One is PV type and the other is PQ type. The PV type is provided the phase angle θ value from the PLL and corresponds to solar power and wind power generators. It plays a role similar to that of the PV bus when power flow calculating. The PQ type is similar to the PV type, but maintains the Q output as a reference value for the purpose of controlling reactive power. It plays a role similar to that of the PQ bus when power flow calculating. Unlike GFL, GFM measures local system data without PLL to form its own phase angle. The main control target simulates the swing equation characteristics of a synchronous generator with voltage and frequency.

B. Synchronization with Grid of GFL

Unlike conventional synchronous generators, GFL is not synchronized with the grid. However, the voltage waveform of the monitoring bus is monitored through the PLL and the voltage phase angle is provided to the high-level control loop. In the inner-current control loop, it is provided the phase angle from the high-level control loop and equalizes it to current

through d-q control, serving as a current source providing current. However, in a weak system, the synchronization ability of the PLL according to a fault is lowered. In a weak system, when a fault occurs, the phase angle change of the voltage before and after the fault becomes larger, so an unstable current is output from the inverter due to the large phase angle difference.

III. SYSTEM STRENGTH INDEX

System strength is defined as the ability of the system to maintain the normal state of the power system to maintain the normal state of the power system in case of system disturbance. The SCR index is a representative index of system strength and is calculated by dividing the short-circuit capacity(MVA) of the connecting bus by the rated output(MW) of the inverter-based resources. The formula of the SCR index is equal to “(1)”. However, the SCR index evaluates only the strength of the bus to which a single inverter-based resource is connected, and does not reflect the interaction between inverter-based resources.

$$SCR_{POI} = \frac{SCMVA_{POI} [MVA]}{P_{dc} [MW]} \quad (1)$$

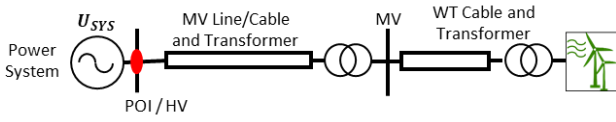


Fig. 1. Concept of SCR index

A. WSCR

ERCOT developed the WSCR index to solve this problem. The WSCR index evaluates system strength by assuming that the interaction of nearby inverter-based resources is 100%. However, since it is assumed to be 100% without considering the actual strength and is an appropriate indicator in special system such as the Panhandle area. The formula of the WSCR index is equal to “(2)”.

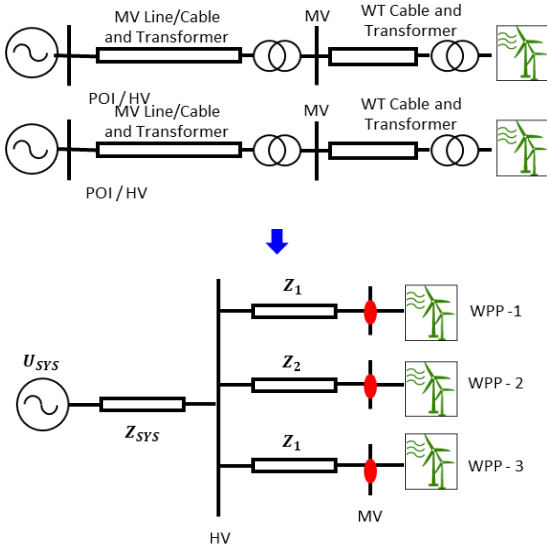


Fig. 2. Concept of WSCR index

$$WSCR_i = \frac{\sum_i^N SCMVA_i P_{dc,i}}{(\sum_i^N P_{dc,i})^2} \quad (2)$$

B. CSCR

GE developed the CSCR index, which is similar to the WSCR index. The CSCR index evaluates the system strength by assuming that the interaction of inverter-based resources connected to the medium-voltage system is 100% and equalizing it with one inverter-based resources. Like the WSCR index, it is evaluated conservatively because it is an interaction assumed to be 100%, not an actual interaction. The formula of the CSCR index is equal to “(3)”.

$$CSCR_i = \frac{SCMVA_i}{\sum_i^N P_{dc,i}} \quad (3)$$

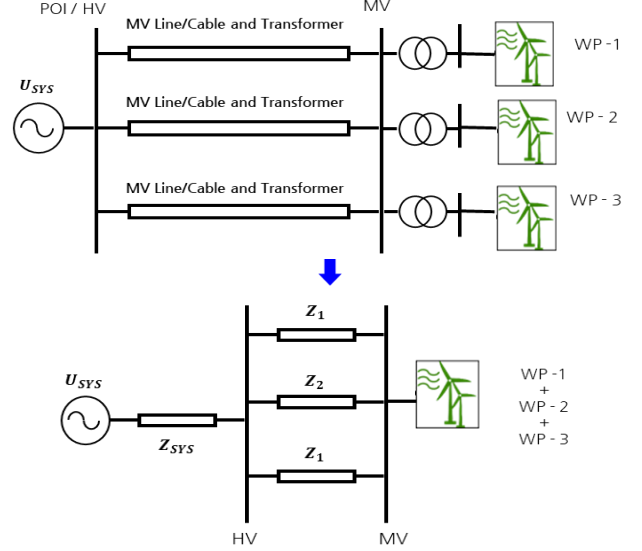


Fig. 3. Concept of CSCR index

IV. CONCLUSION

Renewable energies such as photovoltaic and wind power generators are inverter-based resources that, unlike conventional synchronous generators, provide limited short-circuit current. Limited short-circuit current deteriorates system strength and adversely affects voltage stability after disturbance. Therefore, it is important to evaluate the system strength of regions where renewable energy is intensively connected. In the case of GFL inverters, a PLL is required for phase angle control, and because of this, interaction between inverter-based resources has a negative effect on system strength, and WSCR and CSCR indices have been developed as indices considering interaction. Unlike the GFL, a GFM inverter that can control the phase angle without a PLL can be an alternative. In the future, we plan to study on the system strength index reflecting the dynamic characteristics of the GFM inverter.

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